Run report: the extension (amendment 5), 2026-10-20 to 2026-10-27

Descriptive. What ran and what it recorded. Interpretation is in the study's `README.md` and `REPORT.pdf`.

- **Study:** `research/gaudry_quartic_d4_exponent_20261015`, `PROTOCOL.md` amendments 5, 5a, 5b (amendment 5 and its predictions X1–X10 were committed before any extension run, at `8f15139b2`). Specs `specs/SPEC-<job>.json` generated by `make_specs_ext.sh`; curves `curves.tsv`; queue `jobs_ext.txt` (first order) and `jobs_ext2.txt` (the order run, amendment 5a); runner `run_queue5.sh`; log `sessions/queue.log`; launch notes `sessions/LAUNCH5.txt`.
- **Binary:** the same `ecbench` as the first 72 jobs (sha256 `8627cd29…0aa3`), not rebuilt.
- **Host:** Claude Code cloud container, rustc 1.97.0. The CPU set changed during the run (four CPUs at launch, then one, two, four again) and the container was restarted three times. Every session ran with `--allow-busy`: isolation earned L0 (46 records) or L1 (8) on the index-calculus records. `S` is a count (`priced_by = counted`) and does not depend on wall time, which is descriptive only.
- **Workloads:** `fp4_torsion(p, start seed, 16)`; chain-2 and chain-3 curves from `curves.tsv` for the plain arm and the rho-only runs, the original curves (start seed 1) for the rest; target seeds `2026102000000 + 100000·chain + 10p + t` (plain, new curves), `2026102990000 + 10000·chain + p` (rho-only, 16 targets), and the original `2026101500000 + 10p + t` for the paired and third-target jobs; measurement seed `202610150`.
- **Arms:** `rho-neg`; `ic-d4-w` (`linalg=wiedemann`), `ic-d4-lp2` and `ic-d4-lp1` (`large_primes=2, 1`), the oracle `gaudry-d4:macaulay=26,weighted=1,chains=1,sparsest=1,replay=1,charge_unsolved=1` and base `fghr-y-line:zero=1` throughout.

Runs requested and completed

74 sessions requested, **74 completed, all verified** (`[d]G = Q`): 20 rho-only sessions of 16 targets (320 verified rho runs); 54 sessions with an index-calculus arm and its paired rho record (44 `ic-d4-w`: 40 on the new curves, 4 third targets; 6 `ic-d4-lp2`: `p = 401, 509, 701` ×2; 4 `ic-d4-lp1`: `p = 197, 293` ×2). No job was left unrun; `sessions/NOT_RUN5.txt` does not exist. **Interruptions** (counts are unaffected): the first launch (2026-10-20T06:05Z) lost its jobs to a CPU set that shrank to one CPU, so jobs assigned to missing CPUs ended in two seconds with no session (amendment 5a); the container was restarted between 2026-10-21 and 26, at about 14:47Z on 2026-10-26 and at about 00:45Z on 2026-10-27, each time losing the jobs in flight (three third-target jobs, then three large-prime jobs after 20 minutes, among them); they were rerun. The half-written sessions and failed-launch logs are in `sessions-interrupted/`.

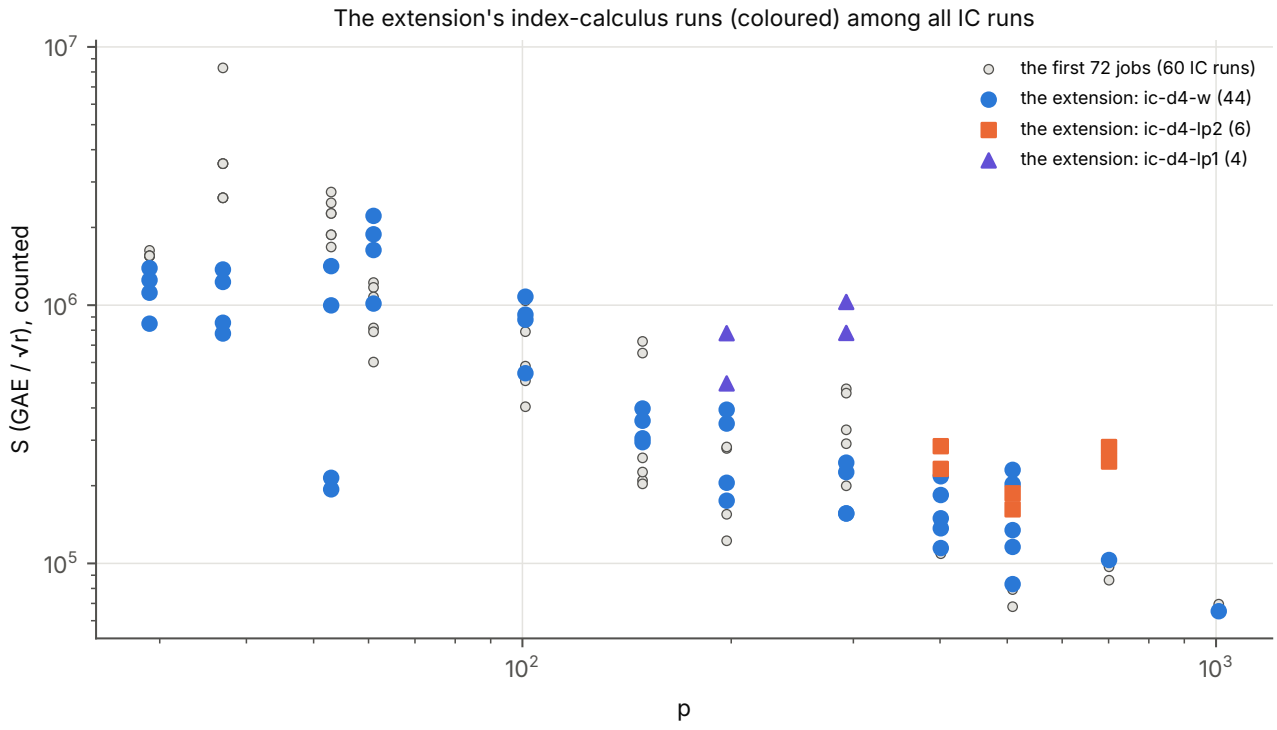


Figure 1: `S` of the extension's 54 index-calculus runs (coloured) among the 60 of the first 72 jobs (grey). Sessions under `sessions/`; summary `raw/summary-ext.json`.

Index-calculus runs (measured; `S` in GAE/√r, trials counted)

job	session	arm	p	status	S	trials	wall h
lp1-p197-t0	ECBS1h095e4c4e7bb9	ic-d4-lp1	197	verified	497757	14689	1.5
lp1-p197-t1	ECBS1ha18b58114155	ic-d4-lp1	197	verified	778876	22098	2.63
lp1-p293-t0	ECBS1h134b2e899887	ic-d4-lp1	293	verified	780716	32618	3.75
lp1-p293-t1	ECBS1h6faad054d102	ic-d4-lp1	293	verified	1028959	42359	5.04
lp2-p401-t0	ECBS1h723d9e95109c	ic-d4-lp2	401	verified	284394	29583	5.97
lp2-p401-t1	ECBS1h26d9d9f039a9	ic-d4-lp2	401	verified	232308	25351	4.07
lp2-p509-t0	ECBS1h454010c656e7	ic-d4-lp2	509	verified	186819	29803	6

lp2-p509-t1	ECBS1hd7a876eb9b63	ic-d4-lp2	509	verified	161977	25175	5.17
lp2-p701-t0	ECBS1h2d7f32eec804	ic-d4-lp2	701	verified	282310	59549	11.86
lp2-p701-t1	ECBS1h8dc37e9be809	ic-d4-lp2	701	verified	247995	56628	10.08
xmain-c2-p29-t0	ECBS1hd15bb4be038f	ic-d4-w	29	verified	1391575	704	0.07
xmain-c2-p29-t1	ECBS1h5af44cf2def8	ic-d4-w	29	verified	1117395	605	0.06
xmain-c3-p29-t0	ECBS1h3214767e6e0c	ic-d4-w	29	verified	847929	325	0.03
xmain-c3-p29-t1	ECBS1h316615ddfd73	ic-d4-w	29	verified	1250979	456	0.05
xmain-c2-p37-t0	ECBS1hcb6182adc976	ic-d4-w	37	verified	854961	469	0.05
xmain-c2-p37-t1	ECBS1hce2bf74eee2e	ic-d4-w	37	verified	776658	447	0.05
xmain-c3-p37-t0	ECBS1hf234a8b6dff3	ic-d4-w	37	verified	1374467	768	0.09
xmain-c3-p37-t1	ECBS1h8d1e234c27b7	ic-d4-w	37	verified	1230078	676	0.08
xmain-c2-p53-t0	ECBS1h3761f6862712	ic-d4-w	53	verified	998201	1650	0.19
xmain-c2-p53-t1	ECBS1h45c754b13a80	ic-d4-w	53	verified	1416242	2283	0.27
xmain-c3-p53-t0	ECBS1hd0ca8a7f03ef	ic-d4-w	53	verified	193541	450	0.05
xmain-c3-p53-t1	ECBS1h7cd7793d301d	ic-d4-w	53	verified	214538	495	0.06
xmain-c2-p61-t0	ECBS1hf0a0eb965baa	ic-d4-w	61	verified	1881357	4191	0.46
xmain-c2-p61-t1	ECBS1h2258b6db08b9	ic-d4-w	61	verified	2217135	4739	0.53
xmain-c3-p61-t0	ECBS1h352e740aa462	ic-d4-w	61	verified	1634861	3591	0.38
xmain-c3-p61-t1	ECBS1h623e9b5ad561	ic-d4-w	61	verified	1014349	2176	0.25
xmain-c2-p101-t0	ECBS1h429c529f8c99	ic-d4-w	101	verified	1078610	4366	0.53
xmain-c2-p101-t1	ECBS1h06353d5fc7c5	ic-d4-w	101	verified	545089	2289	0.25
xmain-c3-p101-t0	ECBS1h3861f87d6169	ic-d4-w	101	verified	878746	4097	0.45
xmain-c3-p101-t1	ECBS1h760c2c2f4a2e	ic-d4-w	101	verified	918952	4329	0.5
xmain-c2-p149-t0	ECBS1hf696d551b8b8	ic-d4-w	149	verified	294713	2916	0.33
xmain-c2-p149-t1	ECBS1hef261382346b	ic-d4-w	149	verified	305088	3045	0.34
xmain-c3-p149-t0	ECBS1h24d6ea93c713	ic-d4-w	149	verified	356851	2987	0.36
xmain-c3-p149-t1	ECBS1h2f6c94aa36e7	ic-d4-w	149	verified	397913	3451	0.39
xmain-c2-p197-t0	ECBS1h89c6a001b6c2	ic-d4-w	197	verified	347945	4924	0.84
xmain-c2-p197-t1	ECBS1hd4be3bde264b	ic-d4-w	197	verified	393867	5805	0.76
xmain-c3-p197-t0	ECBS1hd3c0a2ec068b	ic-d4-w	197	verified	205262	4160	0.53
xmain-c3-p197-t1	ECBS1h3afc5731fa38	ic-d4-w	197	verified	174935	3620	0.49
xmain-c2-p293-t0	ECBS1hb80e36979659	ic-d4-w	293	verified	245612	7230	0.94
xmain-c2-p293-t1	ECBS1h16e11d8378cd	ic-d4-w	293	verified	225659	6632	0.81
xmain-c3-p293-t0	ECBS1h12f4333a28aa	ic-d4-w	293	verified	156201	4691	0.59
xmain-c3-p293-t1	ECBS1h97d38b1e1da5	ic-d4-w	293	verified	156060	4507	0.54
main-p401-t2	ECBS1hb4863eff8cc8	ic-d4-w	401	verified	114658	11770	2.14
xmain-c2-p401-t0	ECBS1he34f7c3deedb	ic-d4-w	401	verified	184006	9543	1.74
xmain-c2-p401-t1	ECBS1h637ec093771f	ic-d4-w	401	verified	217438	10936	2.02
xmain-c3-p401-t0	ECBS1h1401e3f20a0d	ic-d4-w	401	verified	136736	10074	1.85
xmain-c3-p401-t1	ECBS1hf42aa1a28b31	ic-d4-w	401	verified	149541	11393	1.95
main-p509-t2	ECBS1h087cf9517f04	ic-d4-w	509	verified	83135	13263	2.39
xmain-c2-p509-t0	ECBS1h75457aca3afd	ic-d4-w	509	verified	230310	18732	3.28
xmain-c2-p509-t1	ECBS1hf8e21c3d4c78	ic-d4-w	509	verified	203053	15835	3.24
xmain-c3-p509-t0	ECBS1h2487f41fa45e	ic-d4-w	509	verified	115862	12724	2.63
xmain-c3-p509-t1	ECBS1hac97c85cd3ef	ic-d4-w	509	verified	134570	14494	3.04
main-p701-t2	ECBS1h61292cbd5a66	ic-d4-w	701	verified	103082	22019	3.86
main-p1009-t2	ECBS1h7316e394d592	ic-d4-w	1009	verified	65289	19229	3.66

Rho-only runs on the new curves (16 targets each, all verified)

job	session	p	runs	verified	mean S
xrho-c2-p101	ECBS1hc232df41b5dd	101	16	16	1.137
xrho-c2-p149	ECBS1hff48f88f30f8	149	16	16	0.899
xrho-c2-p197	ECBS1ha1dc5acfe9db	197	16	16	0.994
xrho-c2-p29	ECBS1haae2d44bd72d	29	16	16	1.622
xrho-c2-p293	ECBS1he07a17580fc8	293	16	16	0.993
xrho-c2-p37	ECBS1h4f1d405c004c	37	16	16	1.493
xrho-c2-p401	ECBS1hf25af09985ca	401	16	16	0.877
xrho-c2-p509	ECBS1hb90ec3d05d16	509	16	16	1.101
xrho-c2-p53	ECBS1h84fc5894f9c2	53	16	16	1.155
xrho-c2-p61	ECBS1hc2e663bf7d9b	61	16	16	1.165
xrho-c3-p101	ECBS1h67384b0c8790	101	16	16	1.265

xrho-c3-p149	ECBS1h04fc8897d8dd	149	16	16	1.136
xrho-c3-p197	ECBS1hcd9c10af6cec	197	16	16	0.961
xrho-c3-p29	ECBS1h8b1ec8140aa9	29	16	16	1.948
xrho-c3-p293	ECBS1h0a315b15e953	293	16	16	0.898
xrho-c3-p37	ECBS1h28e0e6546886	37	16	16	1.460
xrho-c3-p401	ECBS1h235ac1a8090b	401	16	16	0.767
xrho-c3-p509	ECBS1h8043d7b5183c	509	16	16	0.942
xrho-c3-p53	ECBS1h391bad72a763	53	16	16	1.026
xrho-c3-p61	ECBS1h09573ea9175c	61	16	16	1.287

Raw records

`sessions/<job>/records.jsonl` (plan, host and session files beside them), logs `sessions/<job>.log`; audit receipts `audits/<job>.audit.json`, list `audits/AUDITS.txt`.